DOTE 6635: Artificial Intelligence for Business Research (Spring 2026)

Agentic AI

Renyu (Philip) Zhang

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Where Are We?

- Reasoning LLMs: Backbone for AI agents
- Replication automation, APE, Autoresearch, FARS: LLM Agents have driven paradigm shift in academic research.
- Agent-based modeling (e.g., structural models): An important and long-standing method in business, economics, and social science in general.
- AI agents are indeed impacting real business, from customer services, market research, coding co-pilot, to supply chain robotics.
- Automation ladder: Robustness (tedious digital labor) → collaboration (human interactions) → exploration (creativity, scientific discovery)

From ChatGPT to OpenClaw: AI that talks → AI that acts

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Agenda

- **LLM Agents**
- Skills
- OpenClaw
- Context Engineering

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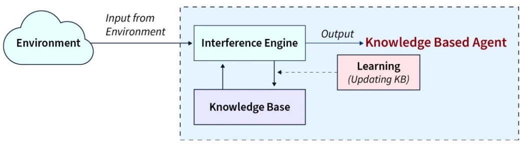
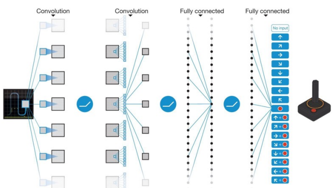
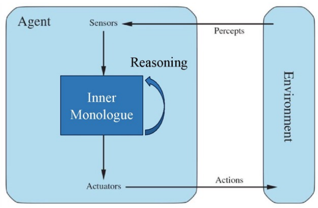
What is Agent?

https://language-agent-tutorial.github.io/slides/I-Introduction.pdf; https://llmagents-learning.org/slides/intro.pdf

- "Definition": An intelligent system that interacts with some environment via a cycle of actions and observations.







	Logical Agent	Neural Agent	Language Agent
Expressiveness	Low bounded by the logical language	Medium anything a (small) NN can encode	High almost anything, esp. verbalizable parts of the world
Reasoning	Logical inferences sound, explicit, rigid	Parametric inferences stochastic, implicit, rigid	Language-based inferences fuzzy, semi-explicit, flexible
Adaptivity	Low bounded by knowledge curation	Medium data-driven but sample inefficient	High strong prior from LLMs + language use

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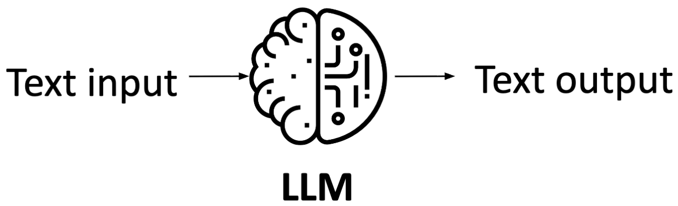
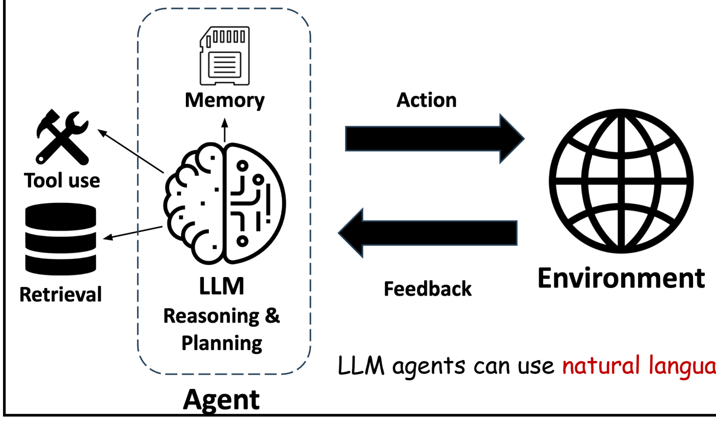
From LLM to Agents

https://language-agent-tutorial.github.io/slides/I-Introduction.pdf; https://llmagents-learning.org/slides/intro.pdf



Brain in a vat



Interactions with environment

LLM agents can use natural language as a vehicle for reasoning and communication.

Conceptual Framework of LLM Agents

https://language-agent-tutorial.github.io/slides/I-Introduction.pdf; https://llmagents-learning.org/slides/intro.pdf

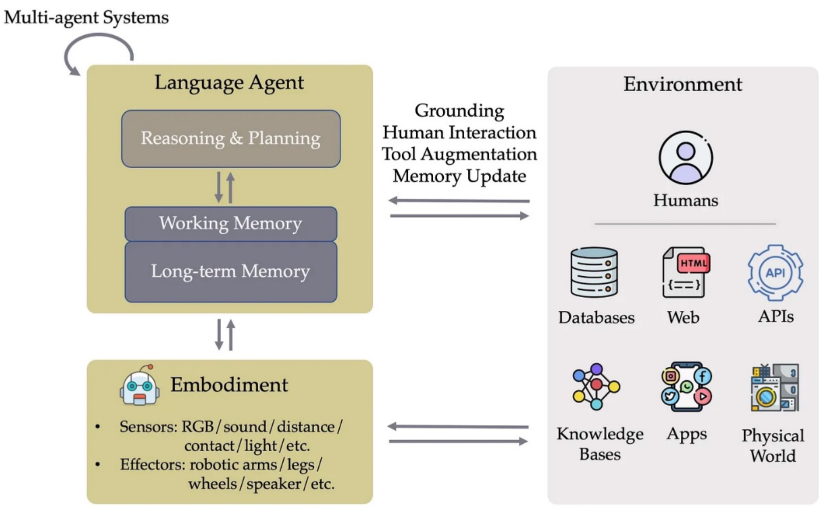


Figure 3. A conceptual framework for language agents. The long-term memory is encoded in the pre-trained parameters and/or an external vector database, while the working memory refers to in-context learning.

- Solving real-world tasks typically involves trial-and-error (a.k.a. agentic loop).
- Leveraging external tools and retrieving from external knowledge expand LLM's capabilities.
- Agentic workflows facilitate complex tasks:
  - Task decomposition
  - Allocation and parallelization of subtasks to specialized modules
  - Division of labor for project collaboration
  - Multi-agent generation inspires better responses

Agenda

- LLM Agents
- **Skills**
- OpenClaw
- Context Engineering

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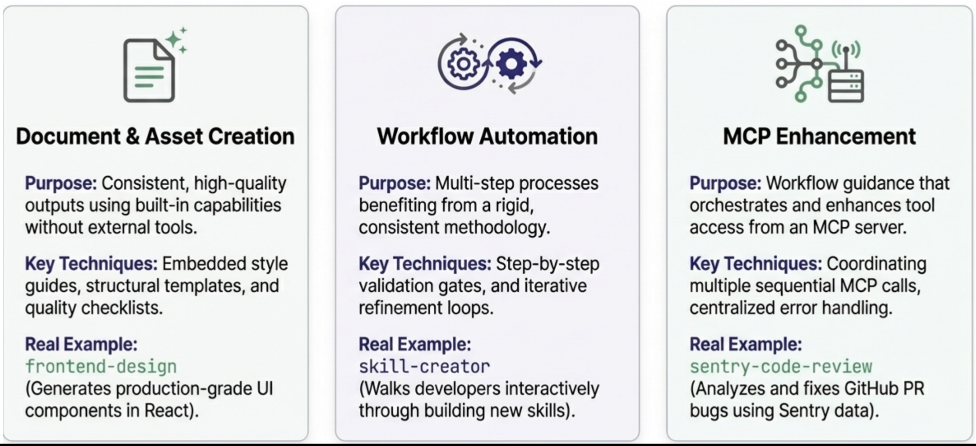
Agent Skills

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf
https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/

- LLM agents are smart and knowledgable PhDs
- Fundametally, skills are SOPs for LLM agents to finish certain tasks.

3 Primary Skills Archetypes



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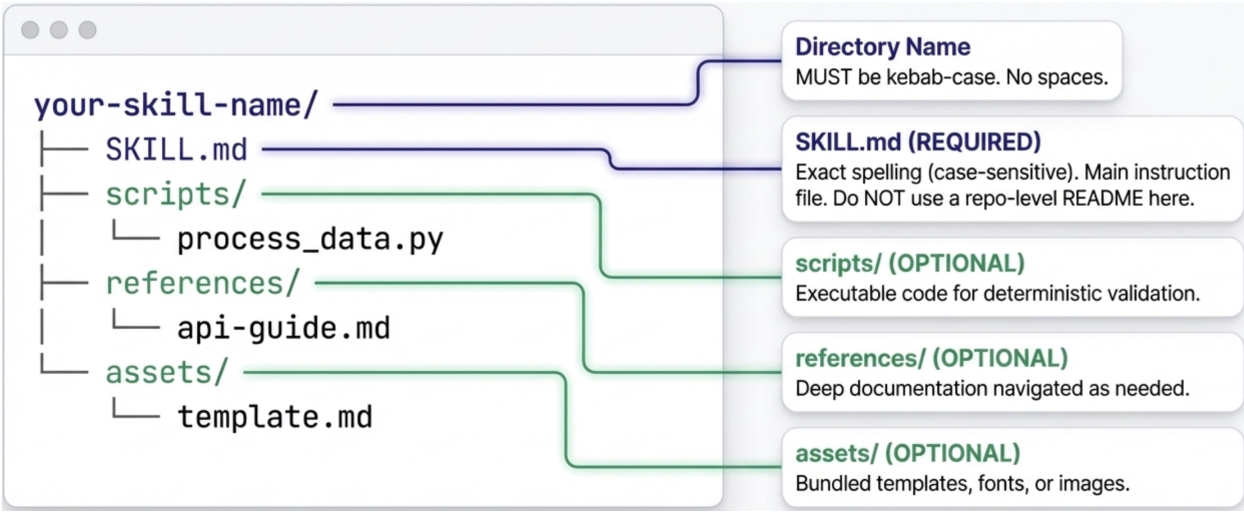
Anatomy of Skills

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf
https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/

- The file architecture may differ, but the logic remains the same for different AI agents.



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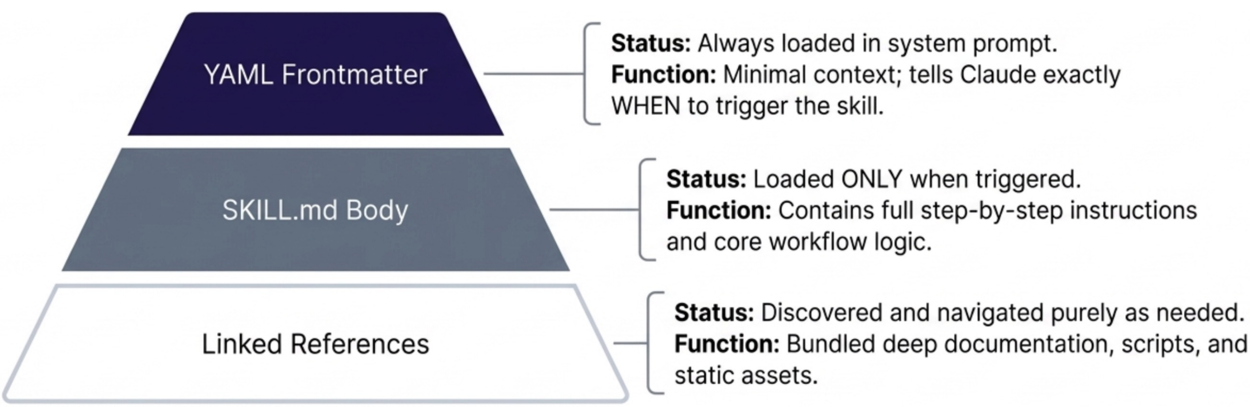
Progressive Disclosure

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf
https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/

- LLM agents should access deep knowledge without easily exhausting context windows.
- Progressive disclosure minimizes token usage while maintaining access to specialized expertise.



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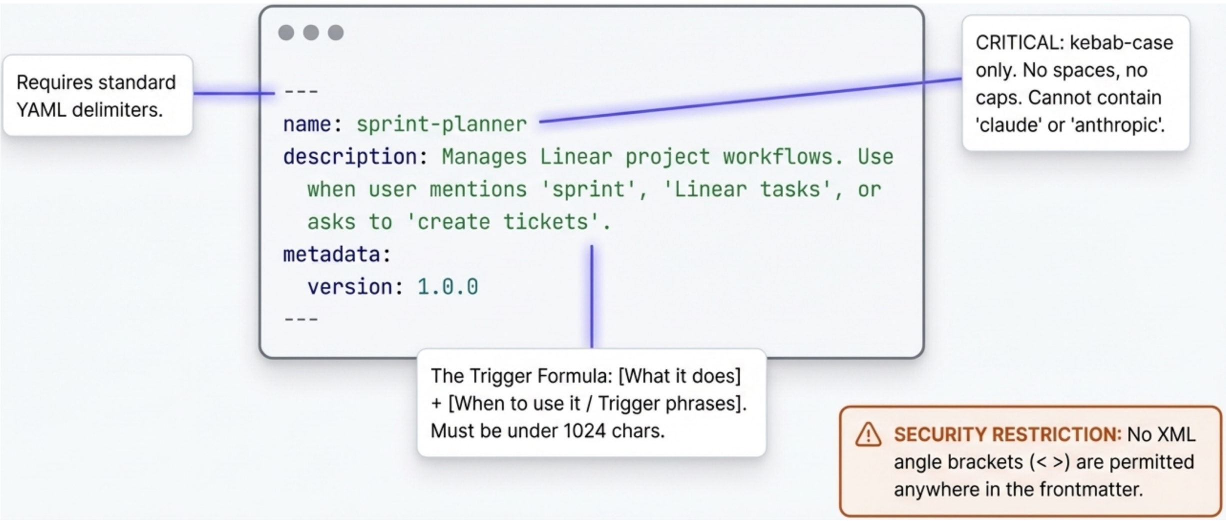
Frontmatter: When the Skill is Triggered

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf
https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/

- This metadata is loaded into the system prompt and determines if your skill is triggered.



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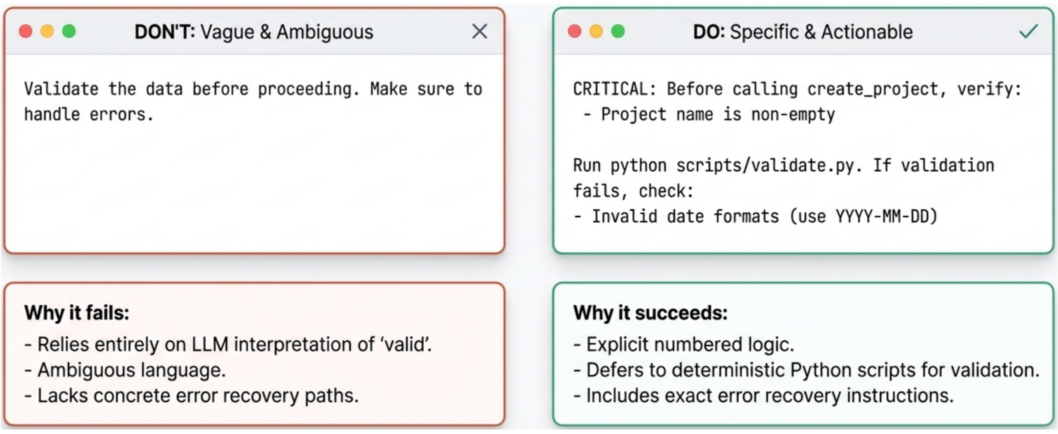
Nonambiguous Instructions

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf
https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/

- Code is deterministic, but natural language is not. Write instructions that leave no room for guesswork.



- You can download useful skills from https://github.com/anthropics/skills or other skills repo/hub and ask your AI agents to write skills for you based on the best practices https://agentskills.io/home.

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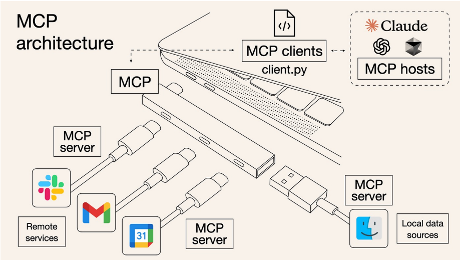
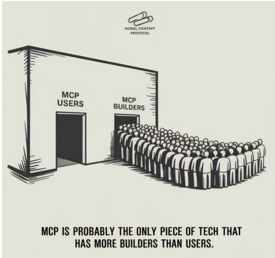
Skills & MCPs

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf
https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/; https://www.anthropic.com/news/model-context-protocol

- Model context protocol (MCP) is a standard for connecting AI agents to apps, data and tools.





MCP (Connectivity)	Skills (Knowledge)
Connects Claude to your service (Notion, Asana, Linear, etc.)	Teaches Claude how to use your service effectively
Provides real-time data access and tool invocation	Captures workflows and best practices
What Claude can do	How Claude should do it

The kitchen analogy

MCP provides the professional kitchen: access to tools, ingredients, and equipment.

Skills provide the recipes: step-by-step instructions on how to create something valuable.

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Agenda

- LLM Agents
- Skills
- <u>**OpenClaw**</u>
- Context Engineering

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OpenClaw

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

- OpenClaw (formerly Clawbot, Moltbot, and Molty) is an open-source agentic system that executes tasks via LLMs using messaging platforms (e.g., WhatsAPP, Lark, Discord) as its main user interface.

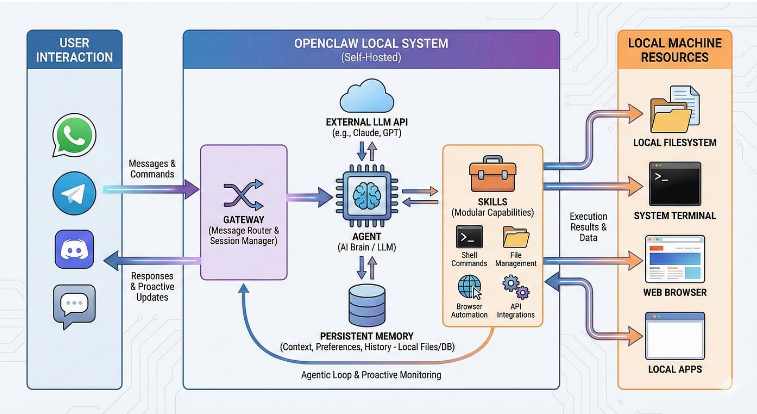
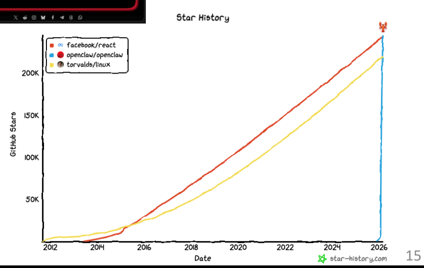


Illustration of OpenClaw Architecture

OpenClaw and Token Economy

https://www.nvidia.com/gtc/keynote/



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Behind OpenClaw's Viral Growth

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

	Consumer Chat (ChatGPT)	Pro Local (Cursor/Codex)	Viral Agentic (OpenClaw)
Capability	Pure Chat & Reasoning	Agentic Loop / File I/O	Agentic Loop / File I/O
Interface	Web UI	IDE / Terminal	WhatsApp, Slack, Lark
Accessibility	Mainstream	Niche (Programmers)	Mainstream

Just as DeepSeek democratized reasoning, OpenClaw democratized the local programming agent by stripping away the IDE and placing it in familiar chat apps.

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Approachability vs. Deep Work

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

Dimension	Chat-Based AI (OpenClaw)	Workspace AI (Cursor/OpenCode)
Workflow Path	Linear & Clunky →	Multi-threaded & Branching
Information Density	Low	High
Observability	Poor (Black Box)	Transparent (Real-time tracking)
Target Audience	Broad Consumer	Deep Knowledge Worker

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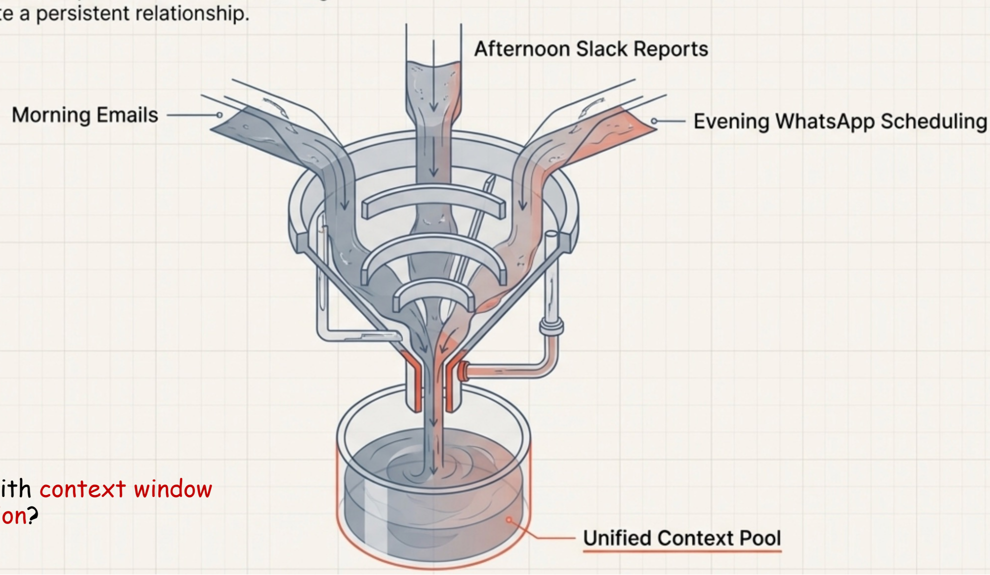
Unified Context

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

Pro tools isolate context. OpenClaw defaults to mixing everything to create a persistent relationship.



How to deal with context window length limitation?

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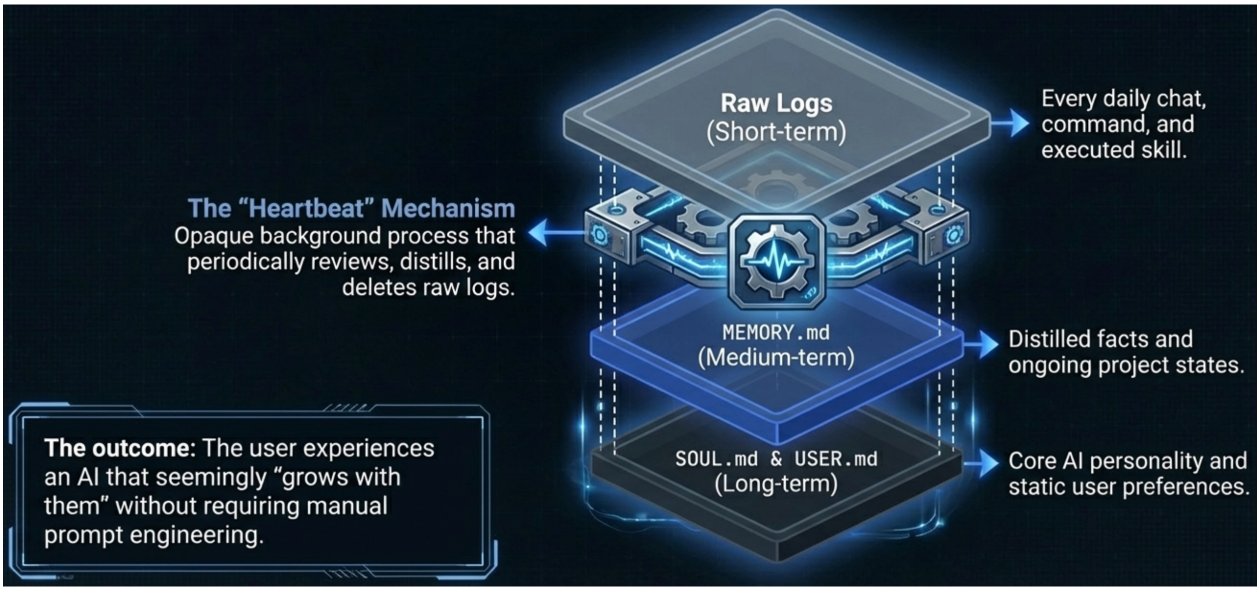
Self-Evolving Memory Engine

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/



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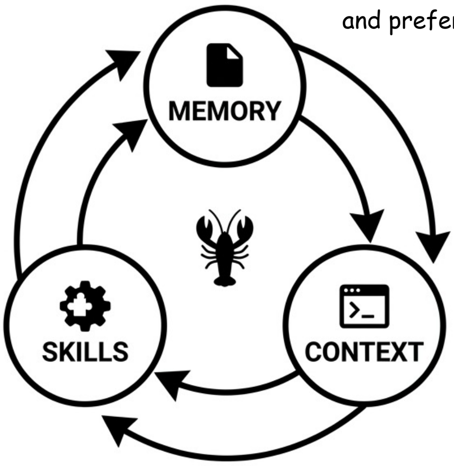
OpenClaw Flywheel

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

Persistent Memory: Retains learnings, habits and preferences over time.

These 3 designs are reinforcing each other, achieving continual learning.



Ecosystem of Skills: Emergent business capabilities and the ability to self-evolve by writing its own code.

Unified Context Pool: Mixes inputs from multiple platforms (WhatsAPP, Telegram, Discord, etc.) into a single data stream.

The AI that actually does things.

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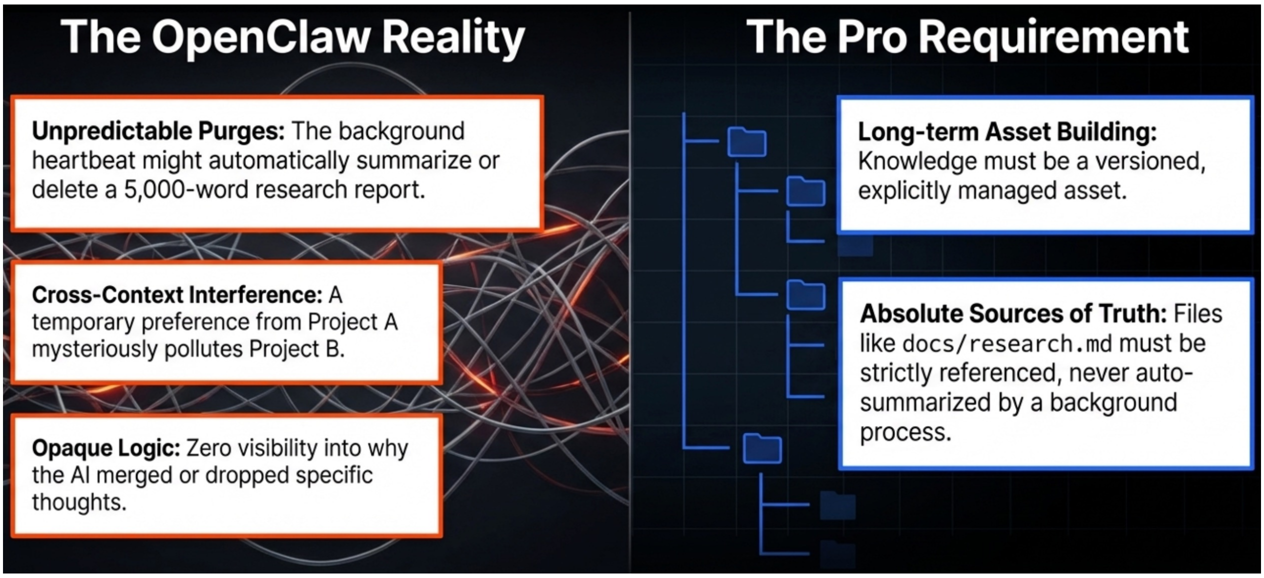
Blackboxed Memory

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/



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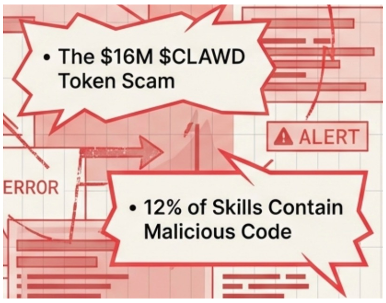
Lethal Trifecta of Agentic AI

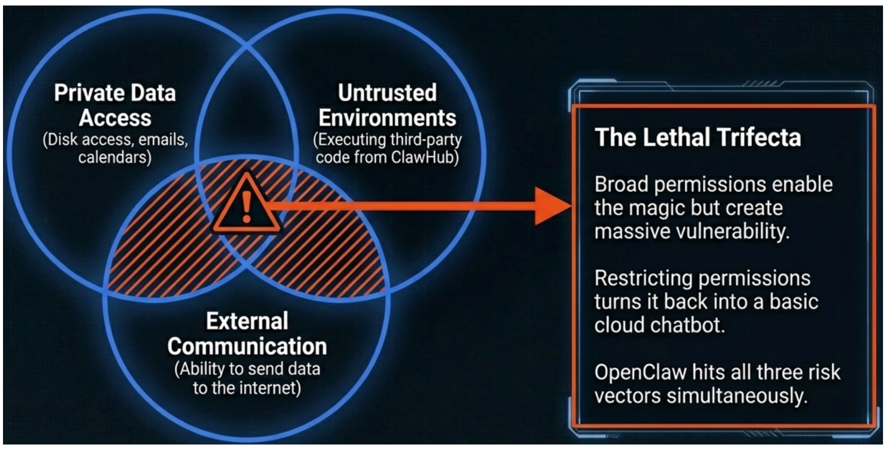
https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/



KOI RESEARCH

ClawHavoc: 341 Malicious Clawed Skills Found by the Bot They Were Targeting

Alex, Oren Yomtov



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Reimplementation Core Architecture

OPENCLAW

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

OpenClaw's viralness demands massive compromises, but we can reconstruct the magic within a controllable architecture.

Actual Needs	Reimplementation
Agentic Loop	OpenCode CLI, or Claude Code, Cursor
Unified Context	Mono Repo
Persistent Memory	Disk-as-Memory (AGENTS.md)
Ubiquity	Custom iOS Client

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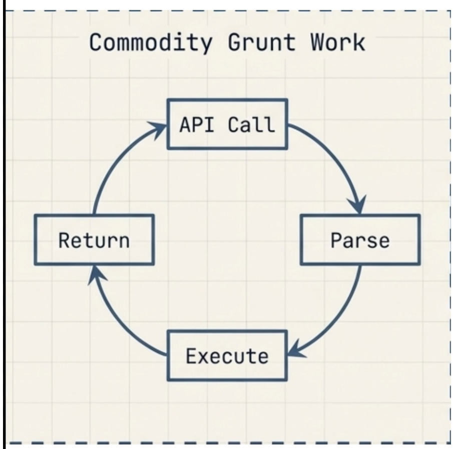
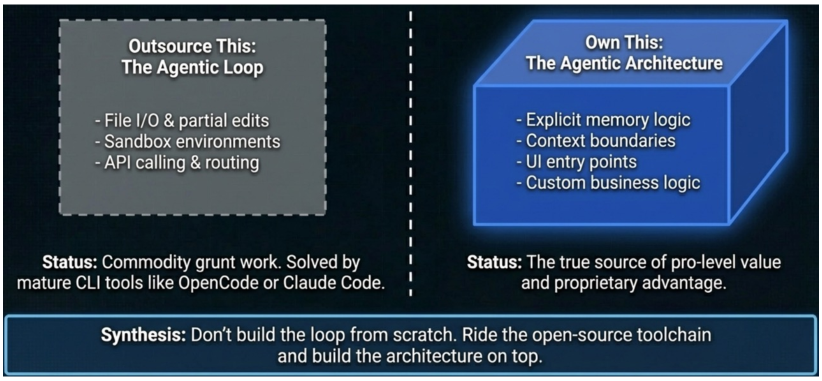
Outsourcing Agentic Loops

OPENCLAW

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/





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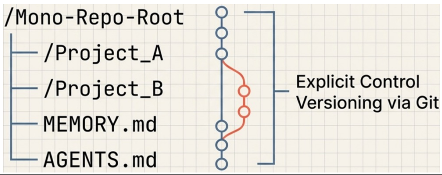
Redesign Memory Architecture

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

Resture fine-grained control over knowledge assets and context using the file system (disk-as-memory) and Git (mono repo).

OpenClaw	Reimplementation
Unit of Memory: Ephemeral database entries.	**Unit of Memory:** Explicit Markdown files (AGENTS.md, MEMORY.md).
Update Logic: Hidden background heartbeat.	**Update Logic:** Transparent, end-of-session persona maintenance workflow.
Control: None. Knowledge cannot be explicitly managed.	**Control:** Total. Tracked via Git, manually editable, with strict file-level rule enforcement.



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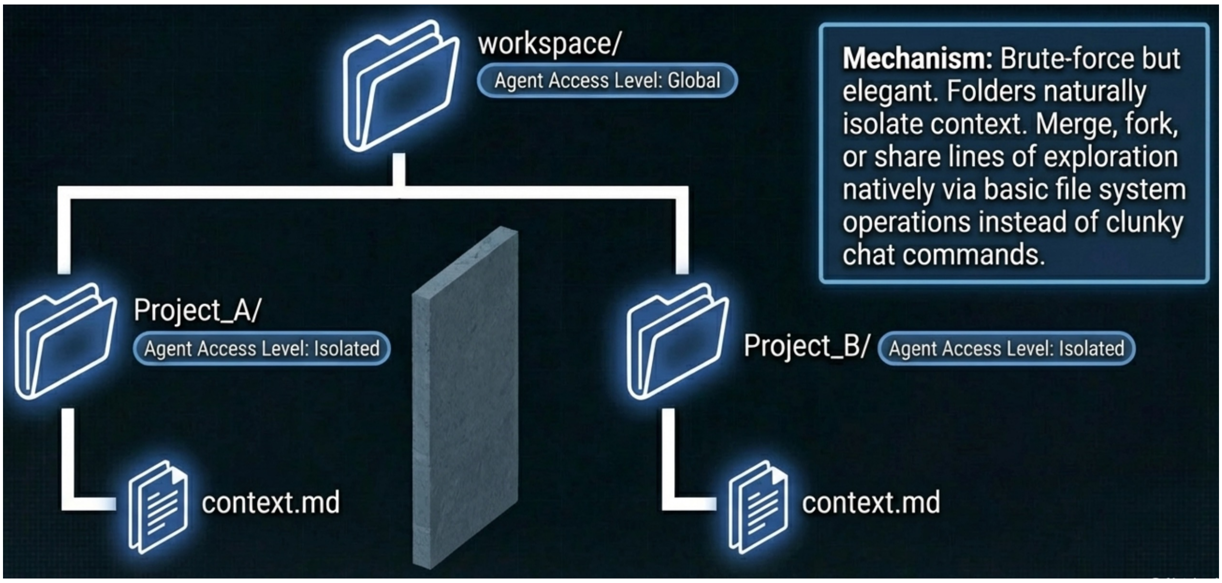
Mono-repo Solves Context Pollution

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/



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One More Thing: Secure the Skills Supply Chain

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf; https://yage.ai/openclaw-en.html; https://openclaw.ai/; https://github.com/openclaw/openclaw; https://clawhub.ai/

Blindly installing 3rd-party MCP servers risks catastrophic supply chain attacks.

It takes only minutes in the age of agentic AI coding, reducing execution risk to zero 0.

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Agenda

- LLM Agents
- Skills
- OpenClaw
- <u>***Context Engineering***</u>

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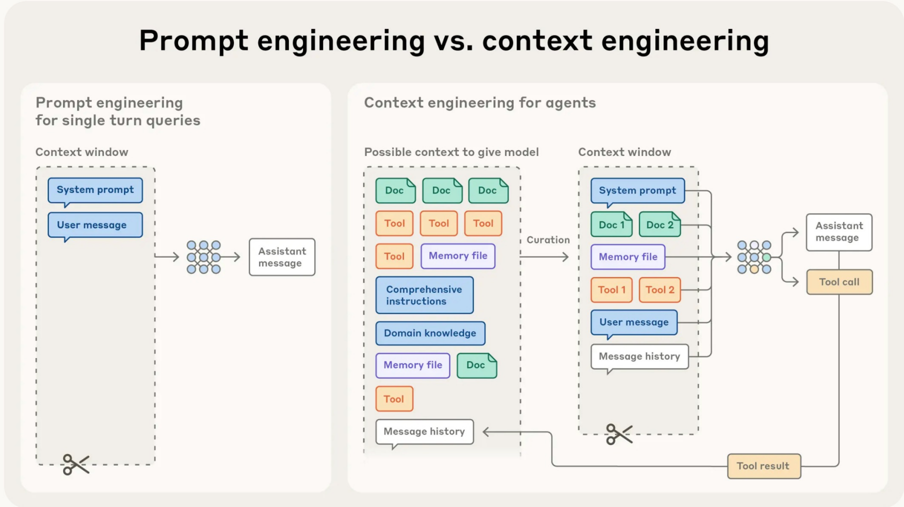
Context is the Key

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

- Context engineering: Curating and maintaining the optimal set of information during LLM inference.



In contrast to the discrete task of writing a prompt, context engineering is iterative and the curation phase happens each time we decide what to pass to the model.

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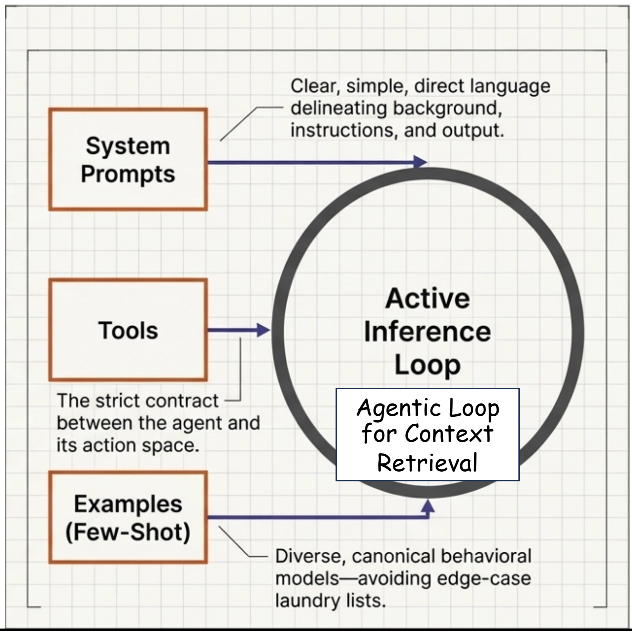
Anatomy of Effective Context

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents



- Anthropic's definition of agents: LLMs autonomously using tools in a loop.
  - Smarter models allow agents to more independently navigate nuanced problem spaces and recover from errors.
- System prompts define the agent attitude.
- Tools define the strict contract between agents and their information/action space.
- Provide examples as few-shot in-context learning.

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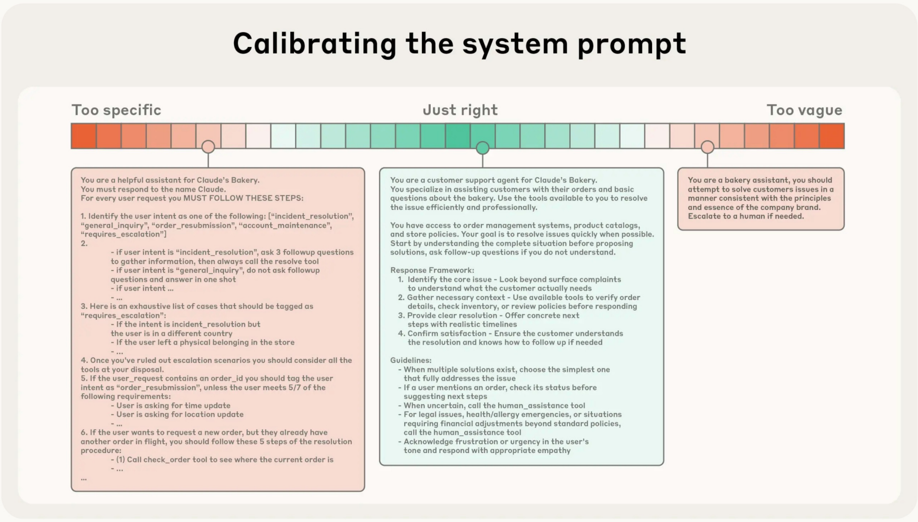
System Prompts

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

- System prompts: The prompt that defines the right altitude for the agent.



At one end of the spectrum, we see brittle if-else hardcoded prompts, and at the other end we see prompts that are overly general or falsely assume shared context.

Organize the prompts into distinct sections like

- <background_info>
- <instructions>
- ### Tool guidance
- ## Output description

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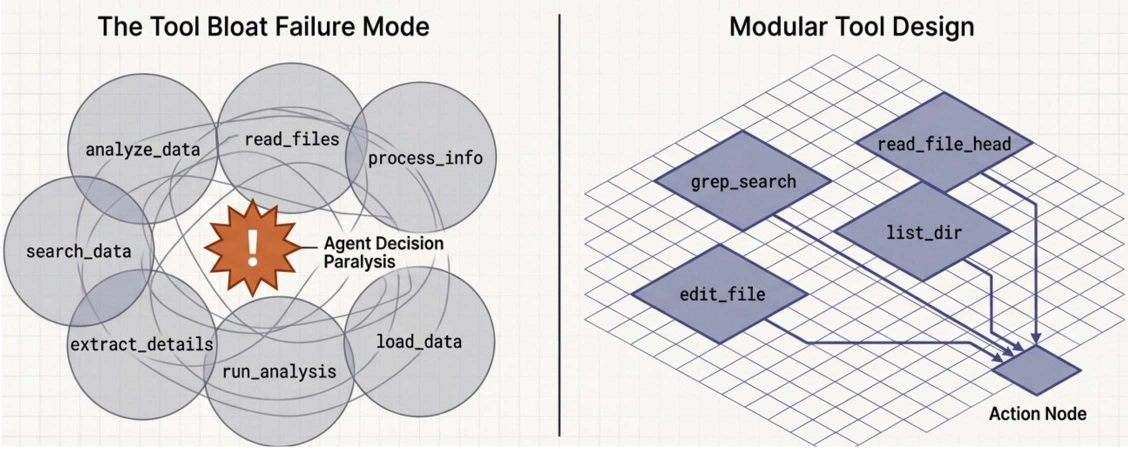
Token-Efficient Action Space

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

Defining a Token-Efficient Action Space

Tools define the strict contract between agents and their environment. A bloated toolset with overlapping functionality guarantees ambiguity and wastes tokens. If a human engineer cannot definitively choose which tool to use in a scenario, an AI agent will also freeze or hallucinate.



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Why Not RAG?

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

The Shift to Agentic Search

While pre-inference retrieval surfaces bulk context upfront, capable agents thrive on Just-In-Time (JIT) retrieval. By maintaining lightweight identifiers instead of raw data, agents bypass stale indexing and preserve their attention budget for active reasoning.

	Pre-Inference Retrieval (RAG)	Just-In-Time (JIT) Agentic Search
Core Mechanism	Embedding-based vector search upfront.	Dynamic loading at runtime via tools.
Token Load	Heavy upfront context injection.	Lightweight references (file paths) in working memory.
Data Freshness	Limited by the last indexing job.	Live, real-time extraction.
Best For...	Static environments (legal, finance).	Highly dynamic codebases and databases.
	Faster	Slower

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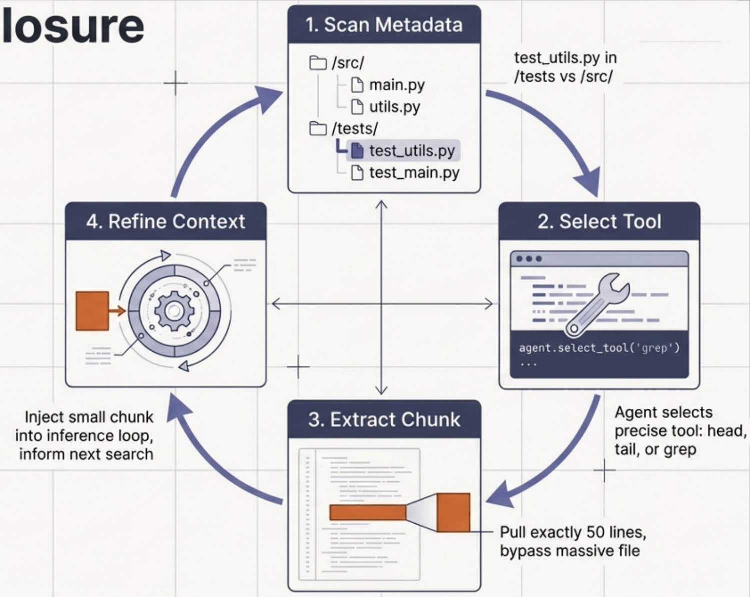
Progressive Information Retrieval

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

Progressive Disclosure

JIT retrieval allows agents to incrementally assemble understanding. Folder hierarchies, naming conventions, and timestamps act as powerful semantic signals.
The agent explores its landscape layer by layer, maintaining only the strictly necessary tokens in working memory.



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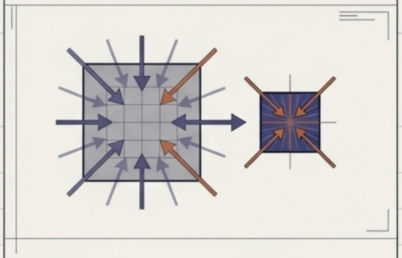
Architecting for Long Horizons

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

For continuous tasks spanning tens of minutes to multiple hours, simply waiting for larger context windows fails due to inevitable context pollution. Sustaining coherence requires architectural interventions that actively prune and persist state outside the primary context window.

Compaction

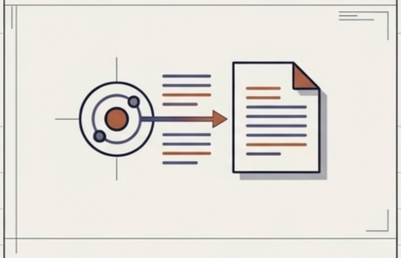


Mechanism: Summarize and reset context.

Overhead: Low.

Best For: Preserving conversational flow and sweeping away dead-ends.

Structured Note-taking

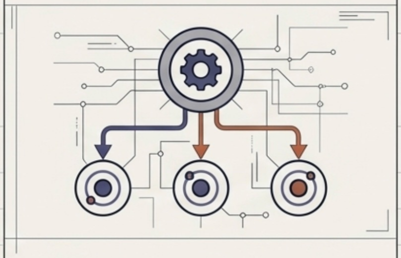


Mechanism: Agent writes to external persistent memory.

Overhead: Medium.

Best For: Iterative development, clear milestones, state-tracking.

Sub-agent Architectures



Mechanism: Specialized agents execute isolated deep-searches.

Overhead: High.

Best For: Complex, parallel research and sprawling codebase migrations.

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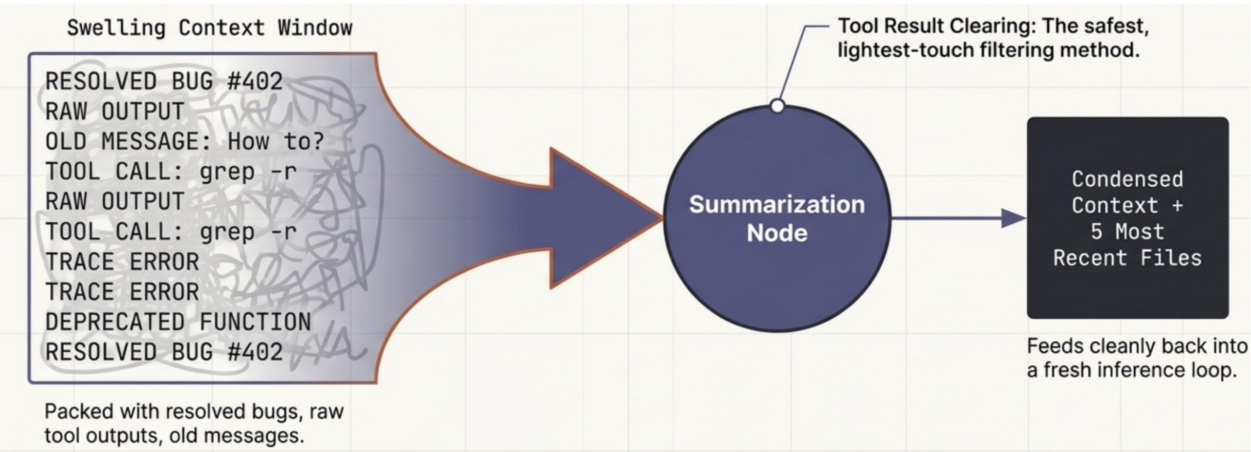
Compaction: Distilling the State

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents



Compaction summarizes a swelling conversation and reinitiates a fresh context window. The art lies in maximizing recall of critical details (architectural decisions, unresolved bugs) while aggressively pruning superfluous content (stale tool calls, redundant outputs).

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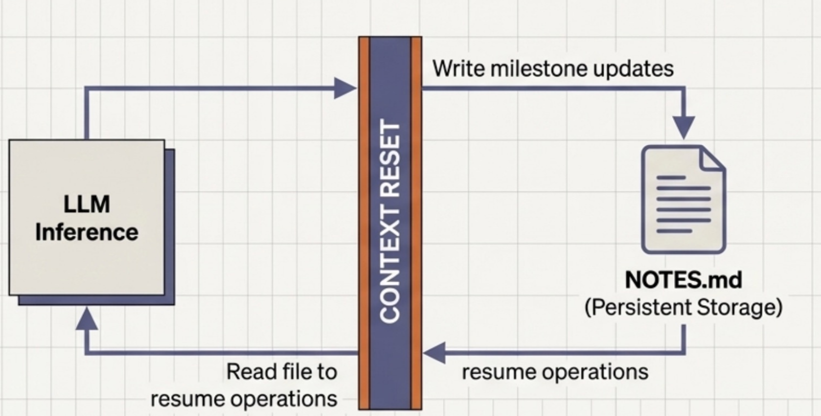
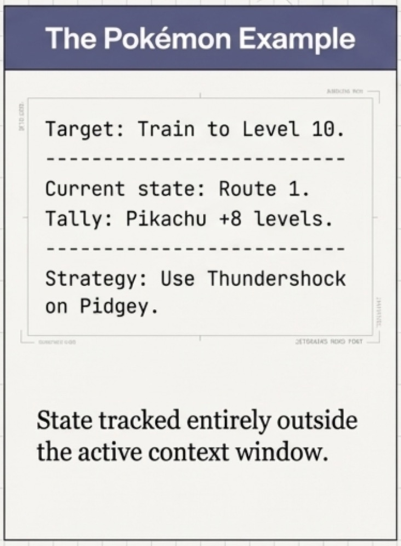
Agentic Memory: Structured Note-Taking

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

By equipping agents with tools to read and write to persistent file systems, they can track complex dependencies across thousands of steps. This simple architectural pattern acts as persistent memory, allowing an agent to maintain strategic coherence across massive context resets.





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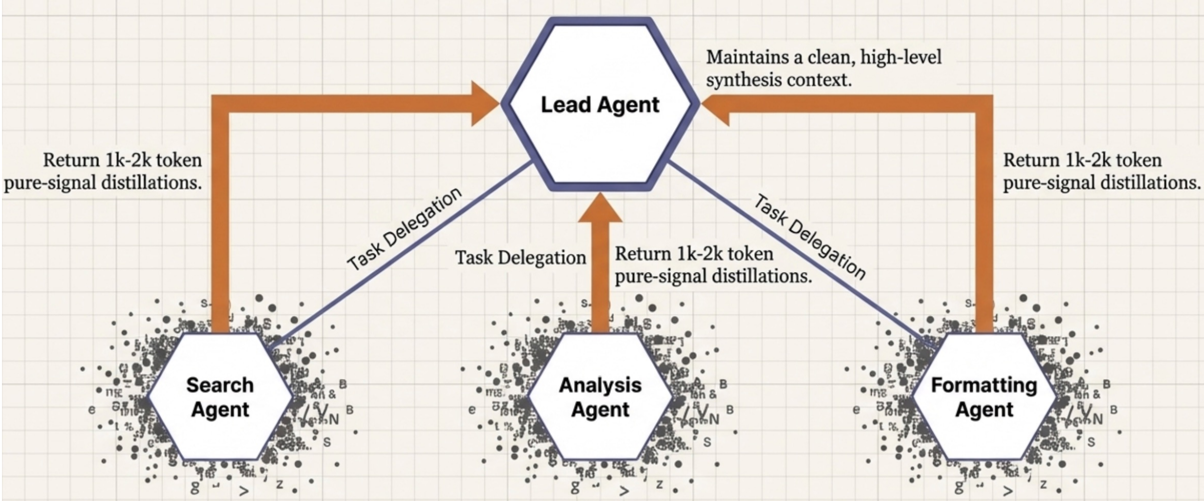
Sub-agent Architectures

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

Rather than forcing one agent to maintain state across an entire project, isolate context pollution. Sub-agents execute deep technical exploration in dirty, high-token environments, but return only highly condensed, pure-signal summaries to the orchestrating lead agent.



Massive, messy context environments (10k+ tokens of raw data).

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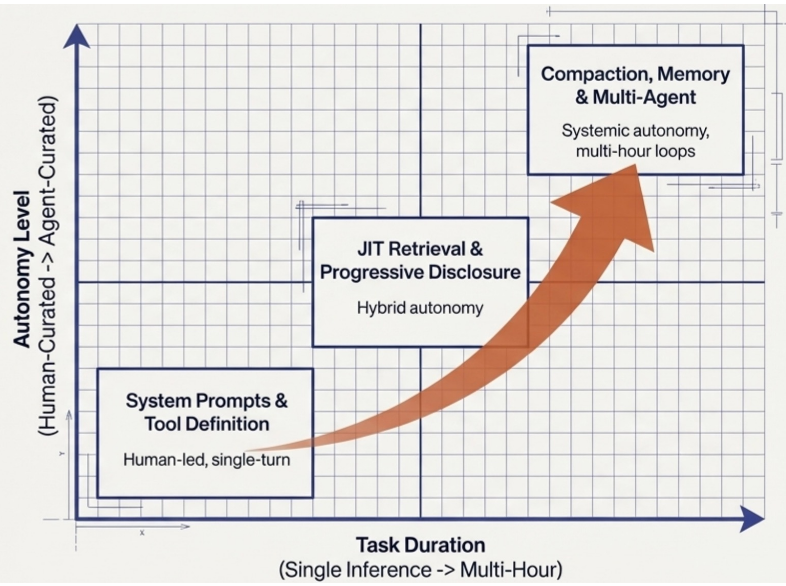
Context Engineering Capability Map

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

The trajectory of AI engineering is clear. As time horizons expand and reasoning capabilities scale, human-curated context must yield to systemic architectures. The ultimate goal is building environments where the LLM dynamically curates its own attention budget.



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Context is a Precious Resource

As underlying models grow smarter, they require less prescriptive hand-holding and demand more architectural freedom. Whether building lightweight single-turn tools or orchestrating multi-hour autonomous clusters, the mandate remains unchanged: engineer the environment to deliver the absolute highest signal with the minimal required tokens.

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents 40

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Context Engineering

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

- Context engineering can by represented in the following abstract format:

$I_1 \leftarrow$ initial input

$C_1 \leftarrow$ empty

For t = 1 to ∞

$$O_t = LLM(I_t, C_t)$$

$$C_{t+1} \leftarrow F(C_t, I_t, O_t)$$

Context window explodes easily.

$I_1 \leftarrow$ initial input

$C_1 \leftarrow$ empty

$C = \{P, M\}$

For t = 1 to ∞

P_t is the part of information delivered to the context of LLM.

$$O_t = LLM(I_t, P_t)$$

$$C_{t+1} \leftarrow F(C_t, I_t, O_t)$$

$$\{P_{t+1}, M_{t+1}\} \quad \{P_t, M_t\}$$

How should we properly design P_t?

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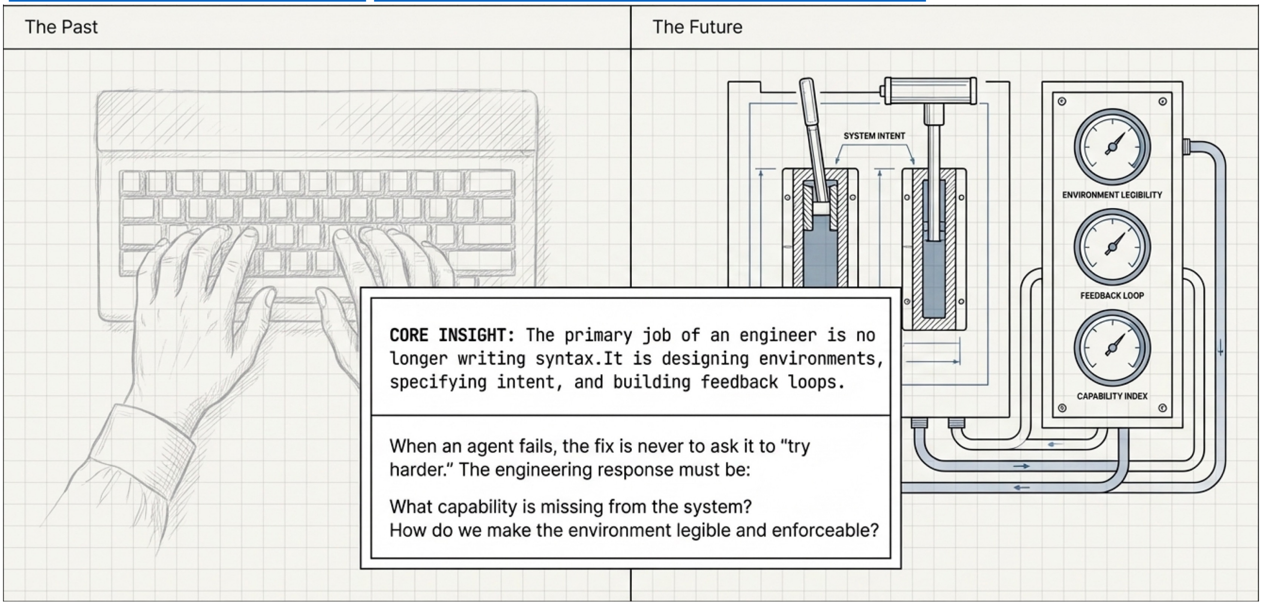
Harness Engineering: Humans Steer. Agents Execute

https://openai.com/index/harness-engineering/; https://www.anthropic.com/engineering/harness-design-long-running-apps



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Engineering (& Research) Paradigm Shift

https://openai.com/index/harness-engineering/; https://www.anthropic.com/engineering/harness-design-long-running-apps
https://github.com/karpathy/autoresearch

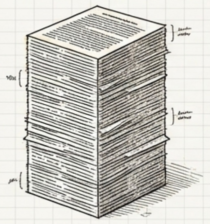
	Traditional	Harness (Agent-First)
What to Do	Writing logic & syntax.	Scaffolding systems & creating leverage.
How to Review & Iterate	Human-to-Human code review.	Agent-to-Agent review; humans review final outcomes.
	Heavy, blocking, manual QA gates.	High-throughput, short-lived PRs; corrections are cheap and fast.
Preference	Optimized for human style and taste.	Optimized for correctness, machine legibility, and invariants.

Recall PS9, auto-research.

Provide Agents a Map, Instead of Manuals

https://openai.com/index/harness-engineering/; https://www.anthropic.com/engineering/harness-design-long-running-apps

- What agents cannot see does not exist.



The Failed Approach: 1,000-Page AGENTS.md

- Crowds out task context.
- When everything is "important," nothing is.
- Rots instantly; becomes an attractive nuisance.

The Winning Approach: 100-Line Table of Contents

- Injected directly into context.
- Acts purely as pointers to deeper, verifiable sources of truth.
- Agents navigate intentionally and explore as needed.

Carefully manage the information structure of your agents!